A RELATIVE ASSOCIATED POINT THAT SPECIALIZES INTO THE FIBRE SPECIALIZES TO AN ASSOCIATED POINT OF THAT FIBRE, UNDER FLATNESS ALONG THE FIBRE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file flat.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the chapter's own retained minimal preamble (sources/stacks-flat/retained-preamble.tex, lines 1-85) is a verbatim excerpt of the native Xy-pic setup and of the native theorem declarations and mathematical macros, followed by the project's own reference/slogan/history display-environment definitions copied mechanically from the shared preamble; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler, and extra_packages is left empty because everything needed is already in the retained preamble. No macro was invented or redefined.

Editorial note. Same-chapter pointers are located at their \label lines inside the pinned chapter file; cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Allowed standard prerequisites (editorial).

- Existence of a generic point of the intersection of the closure of the point with the fibre, providing the intermediate specialization. Standard topology of schemes, used at proof lines 5187-5190. The intersection of $\overline{\{x'\}}$ with the fibre $f^{-1}(\{s\})$ is nonempty and closed in that fibre, so it has a generic point x, giving $x' \leadsto x \leadsto t$.
- Membership in the relative associated points is detected after flat base change. More on Morphisms, label more-morphisms-lemma-associated-point-specializes, invoked at proof line 5191. Applied at the intermediate point x, where $\mathcal{F}$ is flat over S by the hypothesis, it yields $x \in \text{Ass}_{X/S}(\mathcal{F})$.

Source reference. Original lemma 0GSI: sources/stacks-chow/tags.txt, source lines 20933–20933. Navigation pointer for \ref{more-morphisms-lemma-associated-point-specializes} used inside the counted span (line 5192); the argument that consumes it is retained in the span.

Lemma 0.1. *Let $f : X \to S$ be a morphism of schemes of finite type. Let $\mathcal{F}$ be a quasi-coherent $\mathcal{O}_X$-module of finite type. Let $s \in S$. Assume that $\mathcal{F}$ is flat over S at all points of X_s. Let $x' \in Ass_{X/S}(\mathcal{F})$ with $f(x') = s'$ such that $s' \rightsquigarrow s$ is a specialization in S. If x' specializes to a point of X_s, then $x' \rightsquigarrow x$ with $x \in Ass_{X_s}(\mathcal{F}_s)$.*

Proof. Say $x' \rightsquigarrow t$ with $t \in X_s$. Then we can find specializations $x' \rightsquigarrow x \rightsquigarrow t$ with x corresponding to a generic point of an irreducible component of $\overline{\{x'\}} \cap f^{-1}(\{s\})$.

Date: stacks-project:flat.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

By assumption $\mathcal{F}$ is flat over S at x. By More on Morphisms, Lemma 0GSI we see that $x \in \mathrm{Ass}_{X/S}(\mathcal{F})$ as desired. $\square$